

Temperature Indicator for °C, °F, °K and °Ra

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A digital temperature indicator is an immensely useful instrument in the fields of electronics, instrumentation and laboratory use for measurement of temperature. The different types and makes of analogue and digital temperature indicators available in the market are expensive and give temperature readings on a single scale. Presented here is a microcontroller (MCU) based temperature indicator that can display the temperature in four different scales such as Celsius, Fahrenheit, Kelvin

and Rankine. It can measure temperatures from 0°C to 150°C.

Celsius temperature scale is based on 0°C for freezing point and 100°C for boiling point of water, with the interval between the two being divided into 100 equal parts. This scale is used in scientific work everywhere.

Fahrenheit temperature scale is based on 32°F for freezing point and 212°F for boiling point of water, with the interval between the two being divided into 180 equal parts.

Kelvin is defined as the scale of temperature with absolute zero as zero and the triple point of water

as 273.16K. (Triple point of a substance is the temperature and pressure at which the three phases—gas, liquid and solid—of that substance coexist in thermodynamic equilibrium.) On this scale, 0K represents absolute zero, which is the temperature at which molecules of a substance have their lowest possible energy. Many physical laws and formulae can be expressed more simply by using an absolute temperature scale. Accordingly, Kelvin scale has become the international standard for scientific temperature measurement.

Rankine scale measures thermodynamic temperature corresponding to Fahrenheit. Zero in Rankine is the same as zero in Kelvin. A temperature of -459.67°F is equal to 0°R.

Relations between Celsius and other scales are given in the table below.

Circuit and working

Fig. 1 shows the circuit diagram of the temperature indicator. It comprises voltage regulator 7805 (IC1), temperature sensor LM35 (IC2), micro-

Relations Between Celsius and Other Scales

Celsius	Fahrenheit	$F = (C \times 1.8) + 32$
Celsius	Kelvin	$K = C + 273.16$
Celsius	Rankine	$Ra = (C \times 1.8) + 32 + 459.67$

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V voltage regulator
- IC2 - LM35 temperature sensor
- IC3 - PIC16F877A MCU
- D1-D5 - 1N4007 rectifier diode
- LED1 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 680-ohm
- R2, R3 - 1-kilo-ohm
- R4 - 10-kilo-ohm
- VR1 - 10-kilo-ohm potentiometer

Capacitors:

- C1 - 1000µF, 25V electrolytic
- C2 - 0.1µF ceramic disk
- C3, C4 - 22pF ceramic disk
- C5 - 10µF, 16V electrolytic
- C6 - 1µF, 25V electrolytic

Miscellaneous:

- LCD1 - 20×4 LCD
- XTAL1 - 4MHz crystal oscillator
- S1 - Tactile switch
- CON1 - 2-pin connector terminal
- X1 - 230V AC primary to 9V AC, 500mA secondary transformer

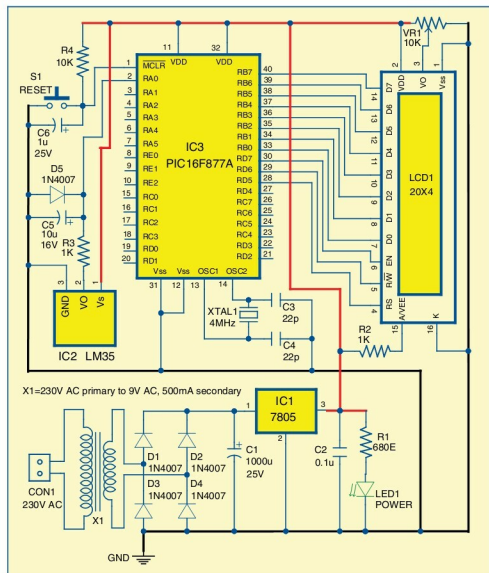


Fig. 1: Circuit diagram of the temperature indicator